

Q1.

A simple electric circuit is a closed path for current to flow. The battery gives the potential difference which pushes the current through the circuit. The switch opens and closes the path. When the switch is closed the current flows through the bulb and the resistor and the bulb glows.

The battery, switch, resistor and bulb and ammeter are joined in series. The ammeter is in series because it measures the current. The voltmeter is connected in parallel across the bulb because it measures the potential difference across it.

If the resistance increases and the voltage stays the same, the current decreases. This is Ohm's law, $V = IR$